THAKUR COLLEGE OF ENGINEERING & TECHNOLOGY
Autonomous College Affiliated to University of Mumbai
Approved by All India Council for Technical Education (AICTE) and Government of Maharashtra (GoM)
Estd. 2001

END SEMESTER EXAMINATION, MAY 2025

T.E. SEMESTER VI (CBCGS-HME 2023)

Branch:	**COMP**	**Q.P. Code:**	**E655RM252-1**
Subject:	**System Programming and Compiler Construction**	**Duration:**	**2 hours**
Subject Code:	**PCC-CS601**	**Max. Marks:**	**60**

Instructions: **1. All sections are compulsory**
2. Figures to the right indicate full marks.

Section-I	**Short Answer Questions (Answer any 05 questions out of 06) (Fundamental, Core Types)**	**(10 Marks)**			
Q. No.		**Marks**	**CO**	**RBT Level**	**PI**
1	Compare Software , Hardware, Firmware with suitable examples.	2	1	AN	3.1.1
2	Define the terms Token, Pattern and Lexeme with examples.	2	2	U	1.3.1
3	Compare SLR, LALR and Canonical LR parsers.	2	3	AN	3.1.1
4	Explain loop optimization techniques.	2	4	U	1.3.1
5	Differentiate between macros and procedure.	2	5	AN	3.1.1
6	Differentiate between absolute and relocatable loader.	2	6	AN	3.1.1
Section-II	**Descriptive Answer Questions (Answer any 04 out of 06) (Descriptive, Comprehension Types)**	**(20 Marks)**			
1	Describe the role of parser.	5	1	U	1.3.1
2	Describe Lexical, Syntax, and semantic errors in the compiler with suitable examples.	5	2	U	1.3.1
3	Apply SLR parsing technique and construct SLR parsing table for the following grammar. S-> (S)S S->€	5	3	A	2.1.3
4	Compute Three-Address Code for a=b*-c +b*-c. Determine Quadruples, Triples, and Indirect Triples for implementing Three-Address code.	5	4	AN	3.1.1
5	Explain databases of Pass-1 and Pass-2 assembler.	5	5	U	1.3.1
6	Explain relocation and linking concepts with examples.	5	6	U	1.3.1
Section-III	**Long Answer Question (Answer any 03 out of 05) (Application, Analytical, Evaluation, Design Type)**	**(30 Marks)**			
1	Identify the need to make grammar left factored. Apply the rules to compute left factored grammar for the given G: 1. A -> abB | abC B -> b C -> c 2. S → a / ab / abc / abcd	10	2	A	2.1.3
2	Determine whether the given grammar is LL (1) or not and construct LL (1) parsing table. S-> aBDh B->Cc C-> bC | € D-> EF	10	3	E	2.2.4

	E-> g | € F-> f | €				
3	Explain various machine dependent and independent code optimization techniques with suitable example.	10	4	R	1.1.1
4	List the various features of macros. Explain different pseudo-ops used for conditional macro expansion along with an example.	10	5	U	1.3.1
5	List databases of Pass1 & Pass2 Direct Linking Loader. Draw flowchart of Pass1 DLL.	10	6	U	1.3.1